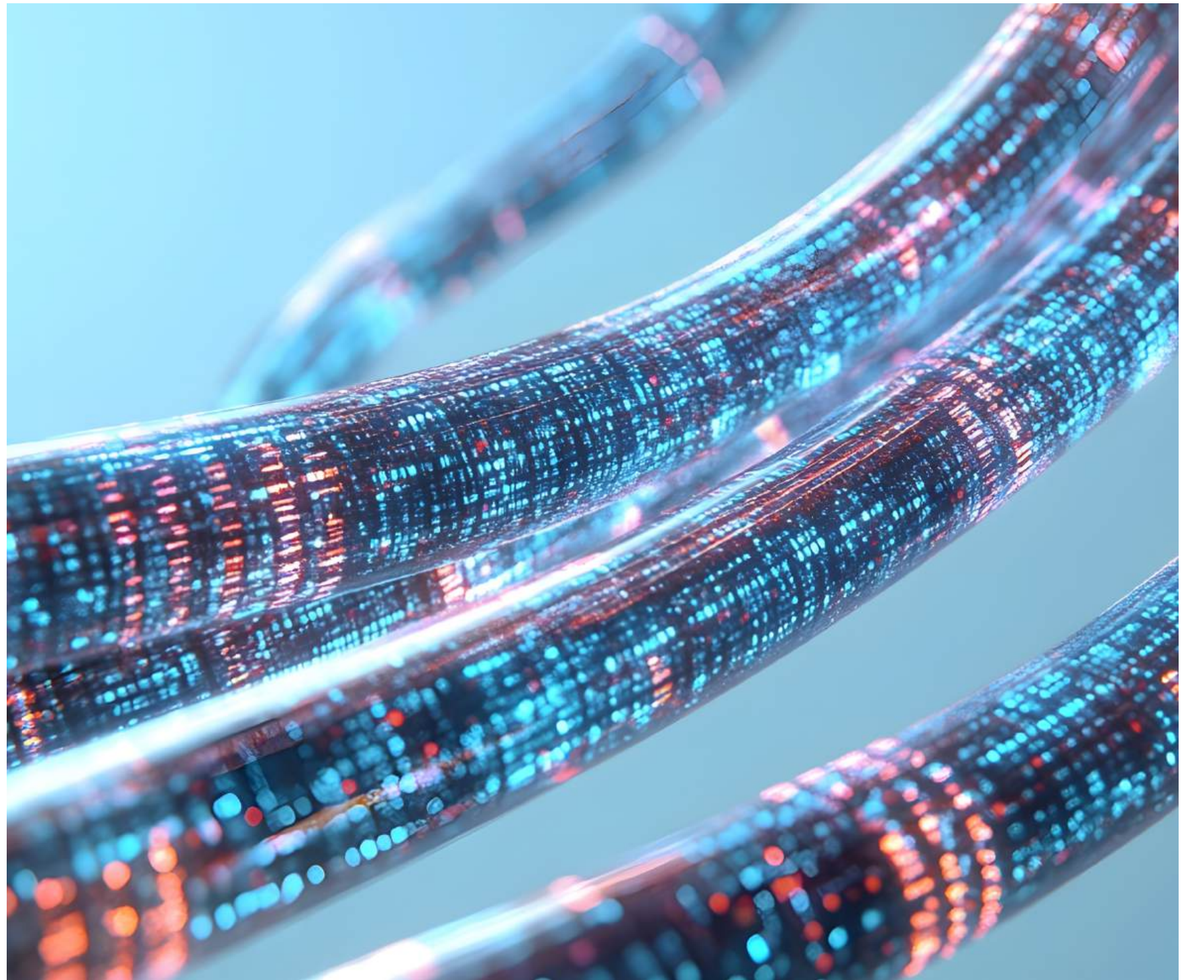
NLP Challenges



NLP SPECIALIST

Introduction to NLP

Understanding the complexities and obstacles in natural language processing today

Let's now take a look at a very, very important and sensitive topic in AI:
Challenges and Ethical Considerations.
We have three broad categories of challenges:

1. **Data-related challenges**
2. **Model-related challenges**
3. **Computational and resource challenges**

1) Data-Related Challenges:

1.1 Bias in Data

AI models are trained using data.
But what if the data itself is biased?
If past hiring data favored **female candidates**, the AI will also favor females.
If past data favored **male candidates**, the AI will favor males.
Biased data = Biased AI decisions.

Data-Related Challenges:

1.2 Data Privacy

Deep learning and large AI models require massive datasets.
If the data comes from users or customers, this creates privacy concerns.
Example:
 Training an AI model using private user information without consent → Major privacy issue.

Data-Related Challenges:

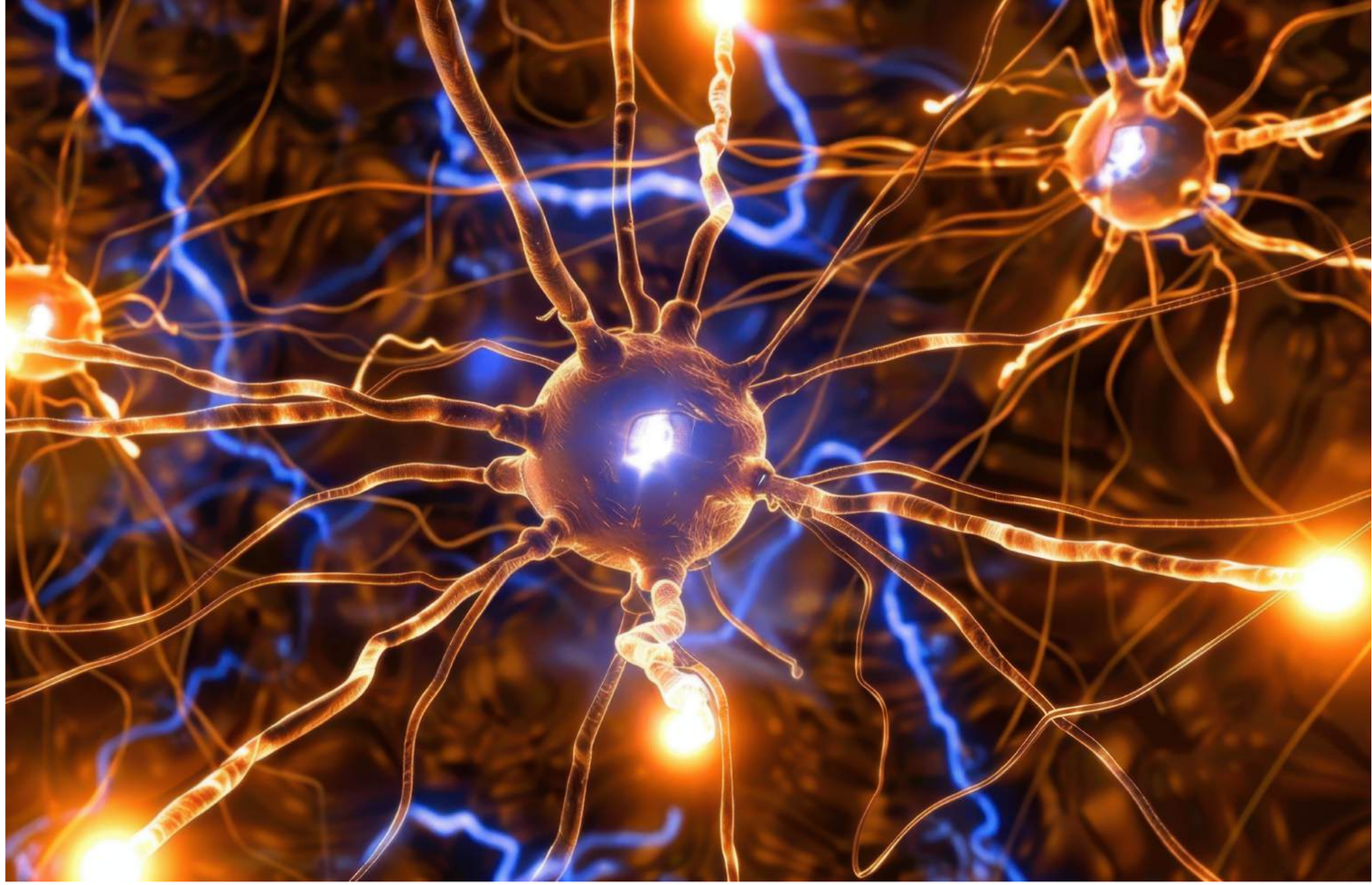
1.3Data Quality

Even if data is large, it may still be:

- incomplete
- unstructured
- low-quality

Example:

A medical AI trained on incomplete patient records → Incorrect diagnosis.



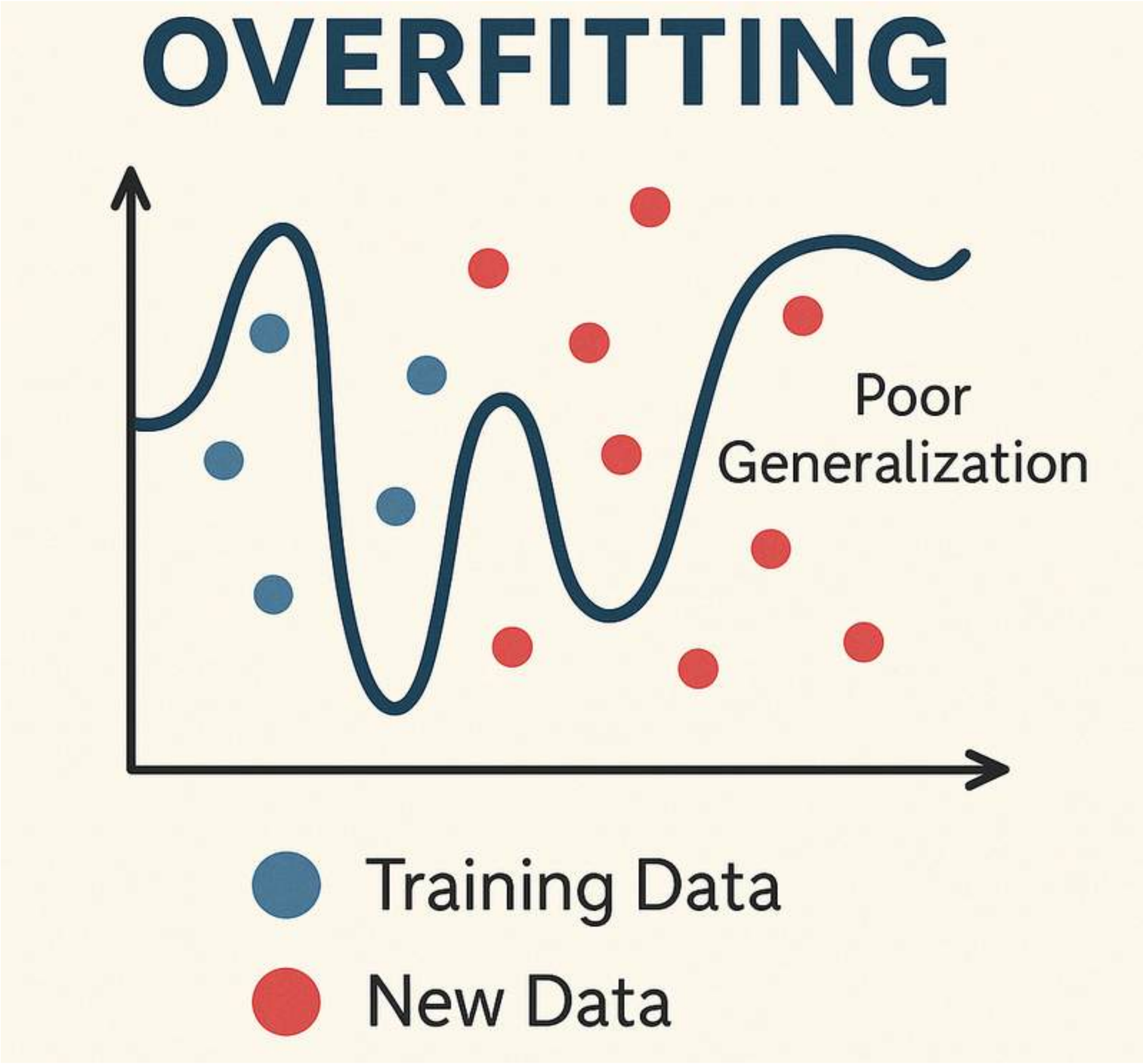
2) Model-Related Challenges:

2.1 Overfitting

In machine learning, models should learn patterns, not memorize data.
If a stock prediction model memorizes past data:

- It works perfectly on known data
- It fails when market conditions change

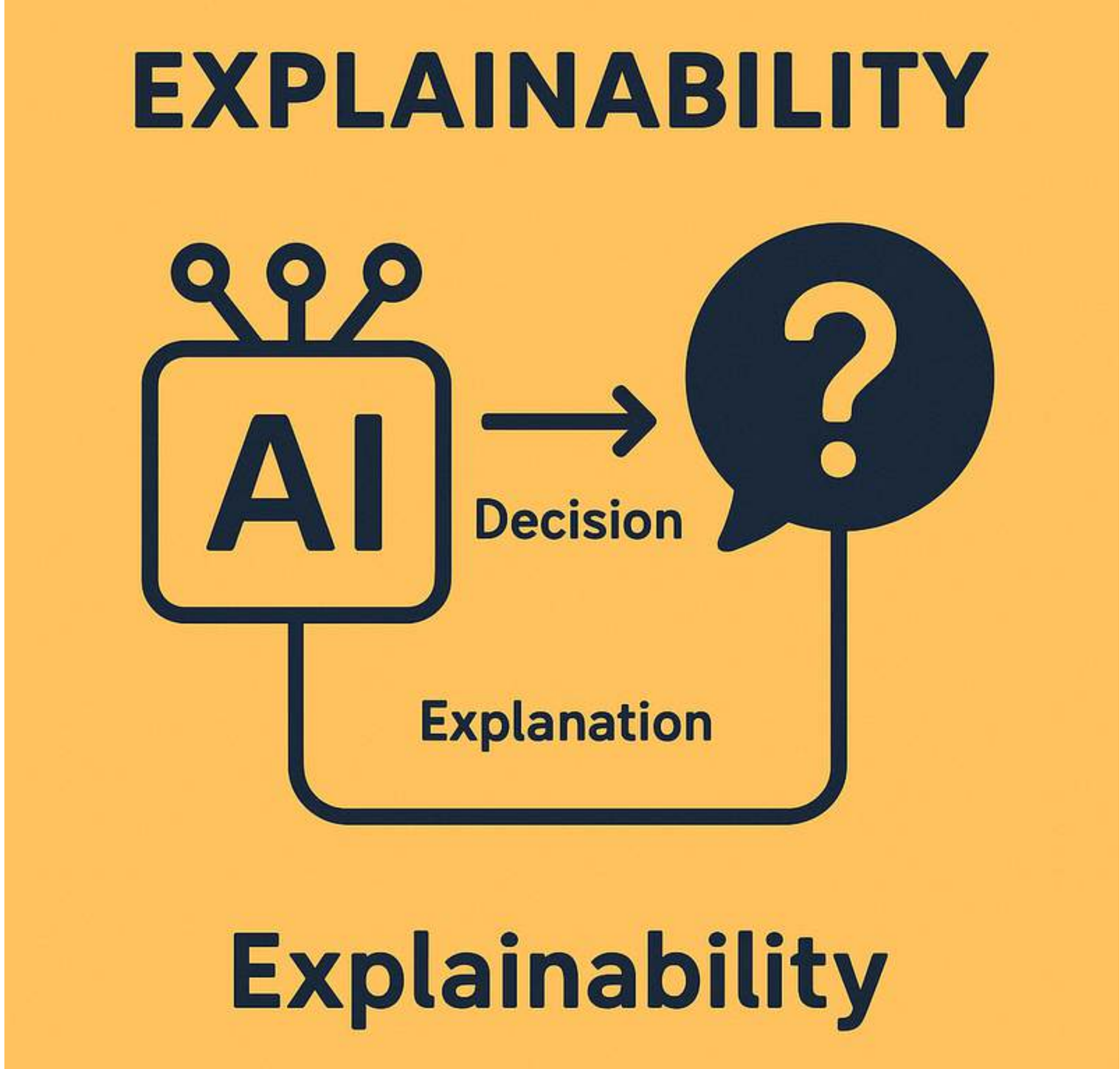
This problem is called overfitting.



Model-Related Challenges:

2.2 Explainability

Deep learning models often act like black boxes.
Example:
A loan approval AI rejects an application but cannot explain why.
This makes it difficult to justify decisions to customers or auditors.



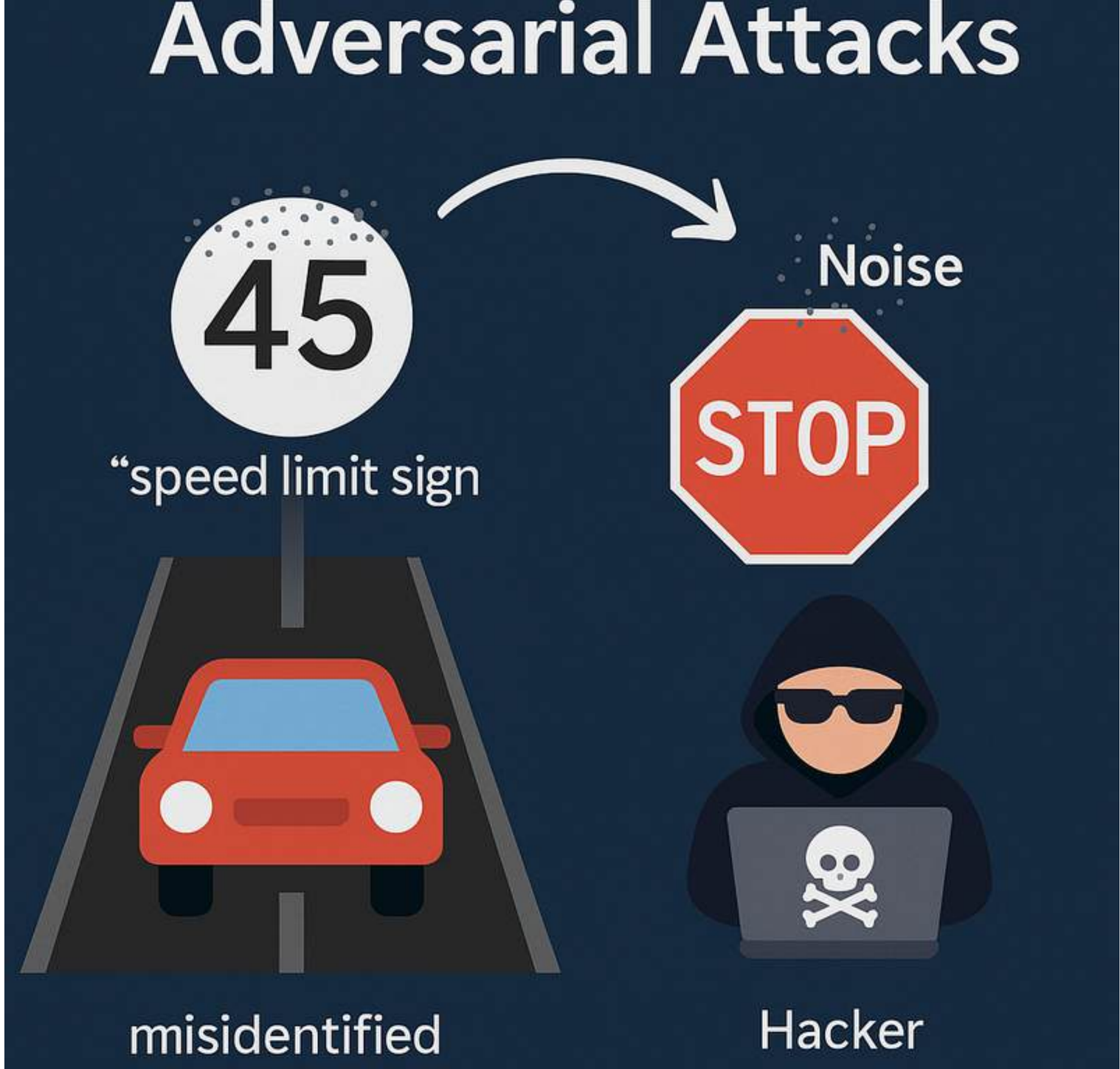
Model-Related Challenges:

2.3 Adversarial Attacks

Hackers can trick AI by slightly changing the input.

Example:

Adding small noise to an image → AI misclassifies a STOP sign as a speed limit sign, which is extremely dangerous for self-driving cars.



3) Computational Challenges

3.1 High Computational Power

Machine learning and deep learning require:

- massive computational power
- expensive GPUs
- huge electricity consumption

Training GPT-4 required thousands of GPUs and massive energy → sustainability concerns.

Computational Challenges:

3.2 Scalability

AI models often fail when moved from lab → real world.

Example:

A chatbot works well with small datasets,

but fails when thousands of customers use it at the same time.

Scalability

Ethical Considerations

Developers must consider:

- Bias & fairness
- Privacy
- Job displacement
- AI safety
- Autonomous decision-making

1) Bias & Fairness

AI can reinforce social inequalities if not trained well.

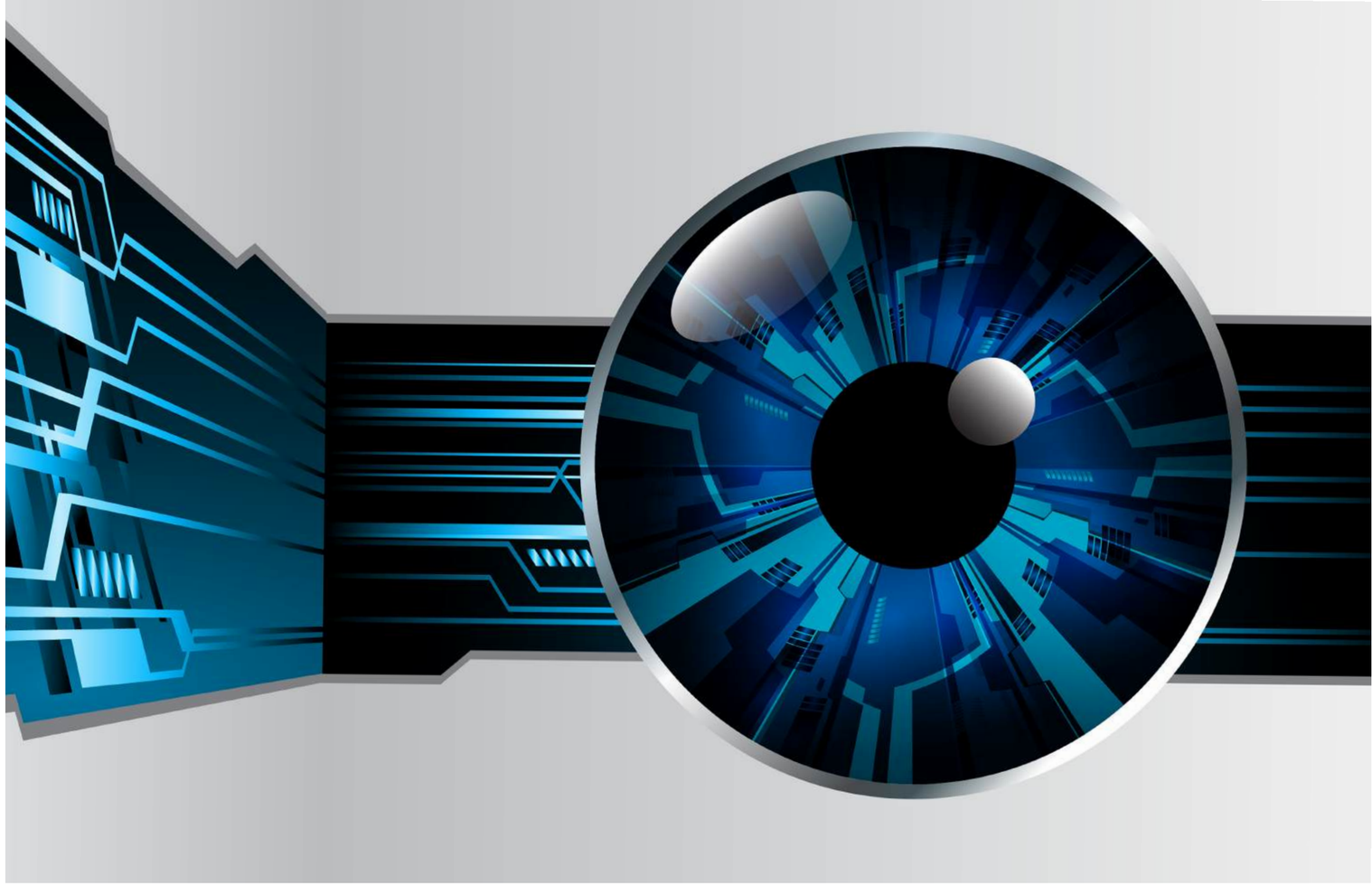
Example:

A predictive policing AI unfairly targets a specific ethnic group.

Solution:

Use diverse data + bias detection tools + fairness-aware training.

2) Privacy



Governments and companies may use AI for mass surveillance.

Example:

China’s Social Credit System monitors citizen behavior using AI.

Solution:

Transparent policies + user consent + limited data collection.

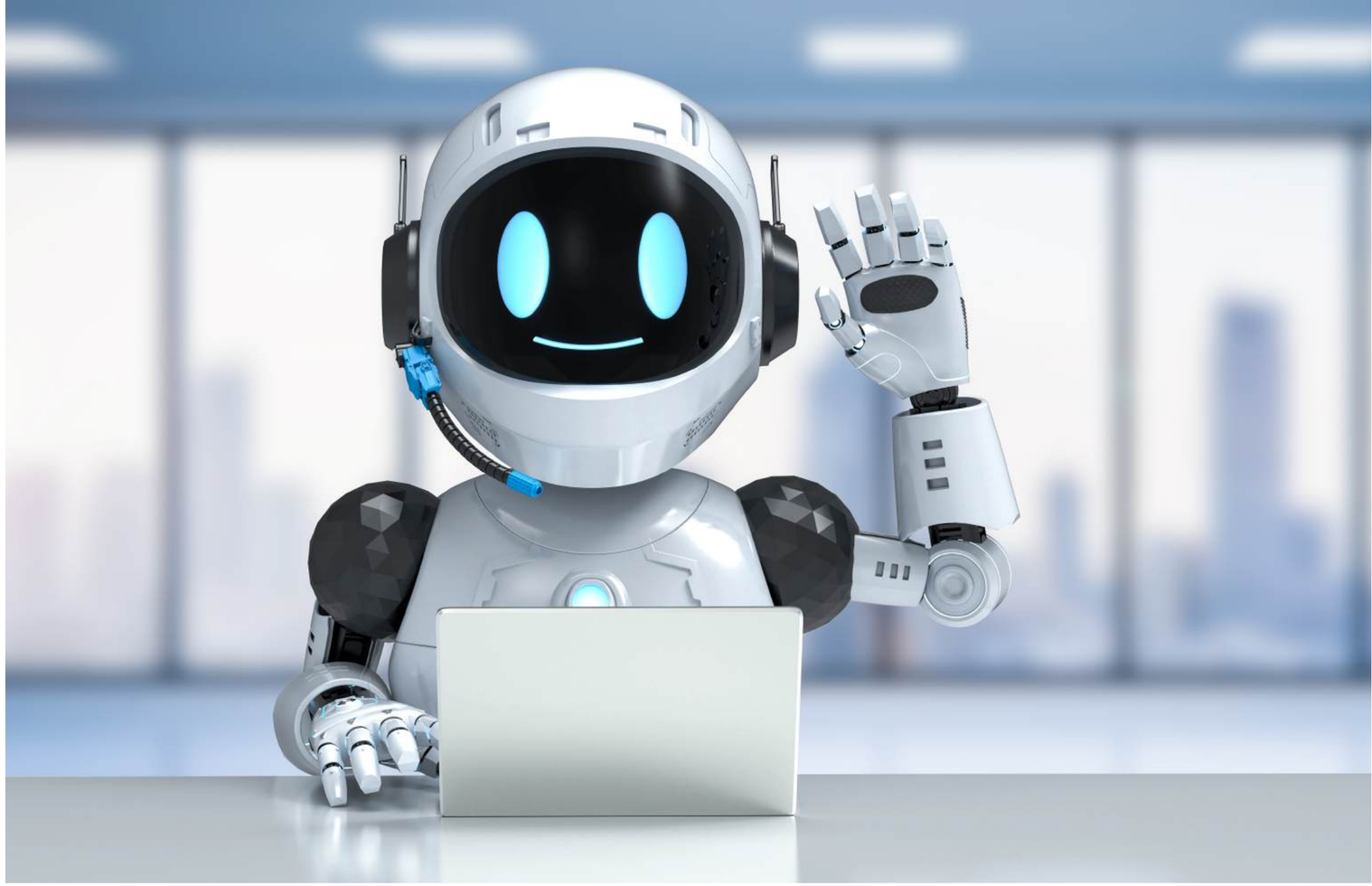
3) Job Displacement & Economy

AI will replace many routine jobs:

- Manufacturing
- Customer service
- Cashiers (self-checkout machines)

Solution:

Reskilling programs + AI education.



4) AI Safety

AI used in life-critical systems (healthcare, transportation) must be safe.

Examples:

- Self-driving cars failing to detect pedestrians → fatal accidents
- Autonomous military drones making strike decisions without humans

Solution:

Strict testing, oversight, and safety protocols.

4) AI Regulations

Governments are introducing AI laws:

- EU AI Act – regulates high-risk AI
- GDPR – protects user data and privacy

Organizations must follow principles:

- Fairness
- Accountability
- Transparency
- Privacy protection